Exercise Sheet 8

Conventions follow Lecture 8. A character of a torus has the form $\exp H \mapsto e^{i\ell(H)}$. Roots satisfy $[H, X] = i\alpha(H)X$ for $H \in \mathfrak{t}$, and $\langle \ell, \alpha^\vee \rangle = \ell(ih_\alpha)$. For a representation Π we use $\Pi(\exp X) = e^{d\Pi(X)}$, as in Lecture 8. Exercise 8.7 is an optional extension.

Exercise 8.1. Let V_m be the space of homogeneous polynomials of degree m in z_1, z_2, with

$$(\Pi_m(g)f)(z) = f(g^{-1}z), \qquad z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \quad g \in \mathrm{SU}(2).$$

Let $r\colon \mathrm{SU}(2) \to \mathrm{SO}(3)$ be a surjective homomorphism with kernel $\{\pm I\}$ that satisfies

$$r\begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix} = R(2\phi), \qquad R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This choice reverses the angle convention of the cover p in Lecture 4. One may take $r(g) = c\, p(g) c^{-1}$, where $c = \mathrm{diag}(1, -1, -1) \in \mathrm{SO}(3)$.

(a) Show that $\Pi_m(-I) = (-1)^m I$. For which m does Π_m descend to $\mathrm{SO}(3)$?

(b) Compute $\Pi_m(\mathrm{diag}(e^{i\phi}, e^{-i\phi}))$ on the monomial $z_1^a z_2^{m-a}$.

(c) Let Π be the representation of $\mathrm{SO}(3)$ obtained from Π_2. Compute $\operatorname{tr} \Pi(R(\theta))$ and compare it with $\operatorname{tr} R(\theta)$.

(d) Decompose $\Pi_2 \otimes \Pi_2$ into irreducible representations of $\mathrm{SO}(3)$. Check dimensions.

Exercise 8.2. In $\mathfrak{so}(3)$ let

$$L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad L_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Let $T = \{\exp \theta L_z\}$ with $\mathfrak{t} = \mathbb{R} L_z$. Use r and $R(\theta)$ from Exercise 8.1.

(a) Compute L_z^2 and L_z^3. Show that $\exp(\theta L_z) = R(\theta)$ and find Λ.

(b) Verify $[L_z, L_x] = L_y$ and $[L_z, L_y] = -L_x$. Show that $E = L_x - iL_y$ satisfies $[L_z, E] = iE$. Find the roots.

(c) Show that $dr(\mathrm{diag}(i, -i)) = 2L_z$. You may use that dr carries the coroot of $\mathrm{SU}(2)$ to the coroot h_α. Conclude that $ih_\alpha = 2L_z$, and check $\alpha(ih_\alpha) = 2$.

(d) Determine $X^*(T)$, P and $X^*(T)^+$. Compare with Exercise 8.1.

Exercise 8.3. Let $\mathfrak{t} = \{\operatorname{diag}(i\theta_1, i\theta_2, i\theta_3) : \theta_1 + \theta_2 + \theta_3 = 0\} \subset \mathfrak{su}(3)$ and $\varepsilon_j(H) = \theta_j$. Let $V(a,b)$ be the irreducible representation of $\mathrm{SU}(3)$ with highest weight $a\varepsilon_1 + b(\varepsilon_1 + \varepsilon_2)$. Its dimension is $\frac{1}{2}(a+1)(b+1)(a+b+2)$. The centre of $\mathrm{SU}(3)$ is $\{I, \omega I, \omega^2 I\}$, where $\omega = e^{2\pi i/3}$.

(a) Show that $H_\omega = \frac{2\pi i}{3}\operatorname{diag}(1,1,-2)$ lies in $\mathfrak{t}$ and that $\exp H_\omega = \omega I$.

(b) Show that ωI acts on $V(a,b)$ by ω^{a+2b}.

(c) Evaluate each root $\varepsilon_j - \varepsilon_k$ on H_ω. Explain why every weight vector gives the same scalar. Verify this on $V(1,0) = \mathbb{C}^3$.

(d) List all (a,b) with $\dim V(a,b) \le 27$ that descend to $\mathrm{PSU}(3)$.

Exercise 8.4. Let $p\colon \mathrm{U}(1) \times \mathrm{SU}(2) \to \mathrm{U}(2)$ be given by $p(z,g) = zg$. For $q \in \mathbb{Z}$ and $m \ge 0$ let $\Pi_{q,m}(z,g) = z^q \Pi_m(g)$ on V_m.

(a) Show that p is surjective with kernel $F = \{(1, I), (-1, -I)\}$.

(b) Explain why the characters of $\mathrm{U}(1)$ are $z \mapsto z^q$ with $q \in \mathbb{Z}$. Show that $\Pi_{q,m}$ descends to $\mathrm{U}(2)$ if and only if $q + m$ is even.

(c) The torus $\operatorname{diag}(e^{i\theta_1}, e^{i\theta_2})$ has characters $e^{i(n_1\theta_1 + n_2\theta_2)}$. Let $q+m$ be even, and let (n_1, n_2) be the highest weight of the representation of $\mathrm{U}(2)$ obtained from $\Pi_{q,m}$. Express (n_1, n_2) through (q, m).

(d) Find (q,m) for $\mathbb{C}^2$, $\det$, the dual $(\mathbb{C}^2)^*$ and $\operatorname{Sym}^k \mathbb{C}^2 \otimes \det^j$.

Exercise 8.5.

(a) Let ρ be the representation of $\mathfrak{u}(1) = i\mathbb{R}$ on $\mathbb{C}^2$ given by

$$\rho(it) = t \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Show that there is no representation Π of $\mathrm{U}(1)$ with $d\Pi = \rho$.

(b) For $s \in \mathbb{C}$ let $\rho_s(it) = ist$ on $\mathbb{C}$. Show that $\rho_s = d\Pi$ for a representation Π of $\mathrm{U}(1)$ if and only if $s \in \mathbb{Z}$.

(c) Let $\mathfrak{gl}(2,\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}(2,\mathbb{C})$ act on V_m, with $\mathfrak{sl}(2,\mathbb{C})$ acting as usual and I acting by $s \in \mathbb{C}$. Show that this module is $d\Pi$ for a representation Π of $\mathrm{U}(2)$ if and only if $s \in \mathbb{Z}$ and $s \equiv m \pmod 2$.

Exercise 8.6.

(a) Let f be central in a compact group K. Suppose f acts by scalars c_1 on V_1 and c_2 on V_2. Show that f acts by $c_1 c_2$ on every subrepresentation of $V_1 \otimes V_2$.

(b) Use the following $\mathrm{SU}(3)$ decompositions:

$$\begin{aligned} (1,0) \otimes (1,0) &= (2,0) \oplus (0,1), \\ (2,0) \otimes (1,0) &= (3,0) \oplus (1,1), \\ (0,1) \otimes (1,0) &= (1,1) \oplus (0,0). \end{aligned}$$

Decompose $(1,0)^{\otimes 3}$. Check dimensions, and check that every constituent descends to $\mathrm{PSU}(3)$.

(c) With the notation of Exercise 8.4, decompose $\Pi_{1,1} \otimes \Pi_{1,1}$ and $\Pi_{1,1} \otimes \Pi_{-1,1}$ into irreducibles of $\mathrm{U}(2)$. Identify the summands.

Exercise 8.7 (Optional extension). For representations Π_m on V_m and Π_n on V_n of $\mathrm{SU}(2)$, let $V_m \boxtimes V_n$ be the representation of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ on $V_m \otimes V_n$ given by

$$(g_1, g_2) \mapsto \Pi_m(g_1) \otimes \Pi_n(g_2).$$

Take the following as given. There is an isomorphism

$$\mathrm{SO}(4) \cong (\mathrm{SU}(2) \times \mathrm{SU}(2))/F_4, \qquad F_4 = \{(I, I), (-I, -I)\}.$$

Under it, $\mathbb{C}^4$ corresponds to $V_1 \boxtimes V_1$. For finite-dimensional vector spaces A and B there is a natural isomorphism

$$\Lambda^2(A \otimes B) \cong (\Lambda^2 A \otimes \mathrm{Sym}^2 B) \oplus (\mathrm{Sym}^2 A \otimes \Lambda^2 B).$$

(a) Show that $V_m \boxtimes V_n$ descends to $\mathrm{SO}(4)$ if and only if $m + n$ is even.
(b) List the irreducible representations of $\mathrm{SO}(4)$ of dimension at most 9.
(c) Decompose $\Lambda^2 \mathbb{C}^4$ and compare with $\dim \mathfrak{so}(4)$.